Lab 4 Submission

Question 1

```
1 / 1    ... a@rsarikaya-VirtualBox: /snap/firefox
1: rsarikaya@rsarikaya-VirtualBox: /snap/firefox
rsarikaya@rsarikaya-VirtualBox:~$ pwd
/home/rsarikaya
rsarikaya@rsarikaya-VirtualBox:~$ cd /usr/share/themes/
rsarikaya@rsarikaya-VirtualBox:/usr/share/themes$ pwd
/usr/share/themes
rsarikaya@rsarikaya-VirtualBox:/usr/share/themes$ cd /
rsarikaya@rsarikaya-VirtualBox:/$ cd/snap/firefox/
bash: cd/snap/firefox/: No such file or directory
rsarikaya@rsarikaya-VirtualBox:/$ cd /snap/firefox
rsarikaya@rsarikaya-VirtualBox:/snap/firefox$
```

```
1 / 1    ...a@rsarikaya-VirtualBox: ~/Documents
1: rsarikaya@rsarikaya-VirtualBox: ~/Documents
rsarikaya@rsarikaya-VirtualBox:~$ cd
rsarikaya@rsarikaya-VirtualBox:~$ pwd
/home/rsarikaya
rsarikaya@rsarikaya-VirtualBox:~$ cd Downloads/
rsarikaya@rsarikaya-VirtualBox:~/Downloads$ cd ../../
rsarikaya@rsarikaya-VirtualBox:/home$ pwd
/home
rsarikaya@rsarikaya-VirtualBox:/home$ ls
rsarikaya
rsarikaya@rsarikaya-VirtualBox:/home$ cd rsarikaya/
.cache/        Documents/     Music/         pictures peru/ Templates/
cis106/        Downloads/     peru/          .pki/          .thunderbird/
cis106.bk/     .gnupg/        picture/       Public/        Videos/
.config/       .local/        pictures/      snap/          .vscode/
Desktop/       .mozilla/      Pictures/      .ssh/          website/
rsarikaya@rsarikaya-VirtualBox:/home$ cd rsarikaya/Documents/
rsarikaya@rsarikaya-VirtualBox:~/Documents$
```

Question 2

```
... aya@rsarikaya-VirtualBox: ~/lab4files
1: rsarikaya@rsarikaya-VirtualBox: ~/lab4files
rw-rw-r-- 1 rsarikaya rsarikaya  284042 Feb 22  2022 csvfile.csv
rw-rw-r-- 1 rsarikaya rsarikaya  764176 Feb 22  2022 document.docx
rw-rw-r-- 1 rsarikaya rsarikaya  742478 Feb 22  2022 globe.avi
rw-rw-r-- 1 rsarikaya rsarikaya      76 Feb 22  2022 helloWorld.c
rw-rw-r-- 1 rsarikaya rsarikaya 2600138 Feb 22  2022 Linux.pdf
rw-rw-r-- 1 rsarikaya rsarikaya    8704 Feb 22  2022 sample.xls
rw-rw-r-- 1 rsarikaya rsarikaya   42777 Feb 22  2022 Tux.png
rw-rw-r-- 1 rsarikaya rsarikaya 2114053 Feb 22  2022 Ubuntu.pdf
rw-rw-r-- 1 rsarikaya rsarikaya  137453 Feb 22  2022 website.html
sarikaya@rsarikaya-VirtualBox:~/lab4files$
```

```
... aya@rsarikaya-VirtualBox: ~/lab4files
1: rsarikaya@rsarikaya-VirtualBox: ~/lab4files
sarikaya@rsarikaya-VirtualBox:~/lab4files$ ls -lh
total 6.5M
rw-rw-r-- 1 rsarikaya rsarikaya  20K Feb 22  2022 countries.json
rw-rw-r-- 1 rsarikaya rsarikaya 278K Feb 22  2022 csvfile.csv
rw-rw-r-- 1 rsarikaya rsarikaya 747K Feb 22  2022 document.docx
rw-rw-r-- 1 rsarikaya rsarikaya 726K Feb 22  2022 globe.avi
rw-rw-r-- 1 rsarikaya rsarikaya   76 Feb 22  2022 helloWorld.c
rw-rw-r-- 1 rsarikaya rsarikaya 2.5M Feb 22  2022 Linux.pdf
rw-rw-r-- 1 rsarikaya rsarikaya 8.5K Feb 22  2022 sample.xls
rw-rw-r-- 1 rsarikaya rsarikaya  42K Feb 22  2022 Tux.png
rw-rw-r-- 1 rsarikaya rsarikaya 2.1M Feb 22  2022 Ubuntu.pdf
rw-rw-r-- 1 rsarikaya rsarikaya 135K Feb 22  2022 website.html
sarikaya@rsarikaya-VirtualBox:~/lab4files$ ls -lhS
total 6.5M
rw-rw-r-- 1 rsarikaya rsarikaya 2.5M Feb 22  2022 Linux.pdf
rw-rw-r-- 1 rsarikaya rsarikaya 2.1M Feb 22  2022 Ubuntu.pdf
rw-rw-r-- 1 rsarikaya rsarikaya 747K Feb 22  2022 document.docx
rw-rw-r-- 1 rsarikaya rsarikaya 726K Feb 22  2022 globe.avi
rw-rw-r-- 1 rsarikaya rsarikaya 278K Feb 22  2022 csvfile.csv
rw-rw-r-- 1 rsarikaya rsarikaya 135K Feb 22  2022 website.html
rw-rw-r-- 1 rsarikaya rsarikaya  42K Feb 22  2022 Tux.png
rw-rw-r-- 1 rsarikaya rsarikaya  20K Feb 22  2022 countries.json
rw-rw-r-- 1 rsarikaya rsarikaya 8.5K Feb 22  2022 sample.xls
rw-rw-r-- 1 rsarikaya rsarikaya   76 Feb 22  2022 helloWorld.c
```

```
Applications  Places
1/1                    Tilix: rsarikaya@rsarikaya-VirtualBox: ~/lab4files
1: rsarikaya@rsarikaya-VirtualBox: ~/lab4files
sarikaya@rsarikaya-VirtualBox:~/lab4files$ ls -lhSgGi
otal 6.5M
99895 -rw-rw-r-- 1 2.5M Feb 22  2022 Linux.pdf
99901 -rw-rw-r-- 1 2.1M Feb 22  2022 Ubuntu.pdf
99904 -rw-rw-r-- 1 747K Feb 22  2022 document.docx
99905 -rw-rw-r-- 1 726K Feb 22  2022 globe.avi
99903 -rw-rw-r-- 1 278K Feb 22  2022 csvfile.csv
99910 -rw-rw-r-- 1 135K Feb 22  2022 website.html
99900 -rw-rw-r-- 1  42K Feb 22  2022 Tux.png
99902 -rw-rw-r-- 1  20K Feb 22  2022 countries.json
99908 -rw-rw-r-- 1 8.5K Feb 22  2022 sample.xls
99906 -rw-rw-r-- 1   76 Feb 22  2022 helloWorld.c
sarikaya@rsarikaya-VirtualBox:~/lab4files$ ls -Xr
ample.xls  Ubuntu.pdf  countries.json  document.docx  helloWorld.c
ux.png     Linux.pdf   website.html    csvfile.csv    globe.avi
sarikaya@rsarikaya-VirtualBox:~/lab4files$
```

Question 3

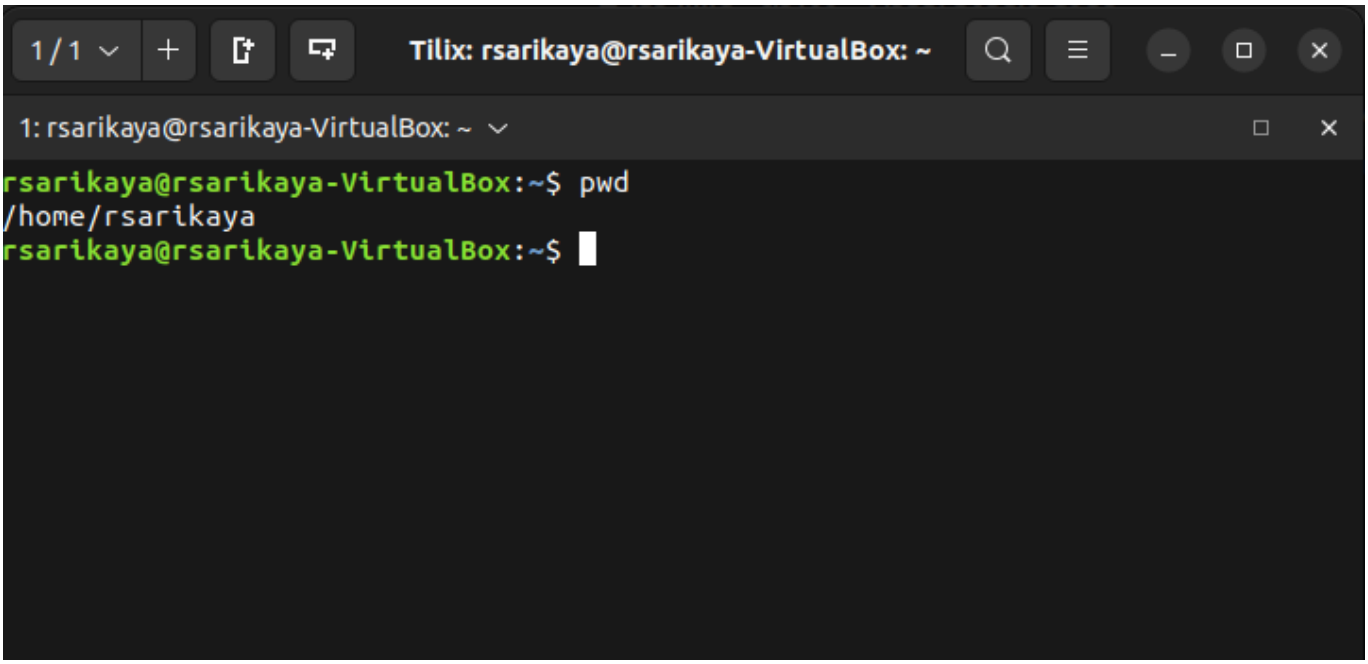
```
rsarikaya@rsarikaya-VirtualBox:~/Downloads$ tree -fph ../lab4files/
[drwxrwxr-x 4.0K]  ../lab4files
├── [-rw-rw-r--  20K]  ../lab4files/countries.json
├── [-rw-rw-r-- 277K]  ../lab4files/csvfile.csv
├── [-rw-rw-r-- 746K]  ../lab4files/document.docx
├── [-rw-rw-r-- 725K]  ../lab4files/globe.avi
├── [-rw-rw-r--   76]  ../lab4files/helloWorld.c
├── [-rw-rw-r-- 2.5M]  ../lab4files/Linux.pdf
├── [-rw-rw-r-- 8.5K]  ../lab4files/sample.xls
├── [-rw-rw-r--  42K]  ../lab4files/Tux.png
├── [-rw-rw-r-- 2.0M]  ../lab4files/Ubuntu.pdf
└── [-rw-rw-r-- 134K]  ../lab4files/website.html

0 directories, 10 files
rsarikaya@rsarikaya-VirtualBox:~/Downloads$
```

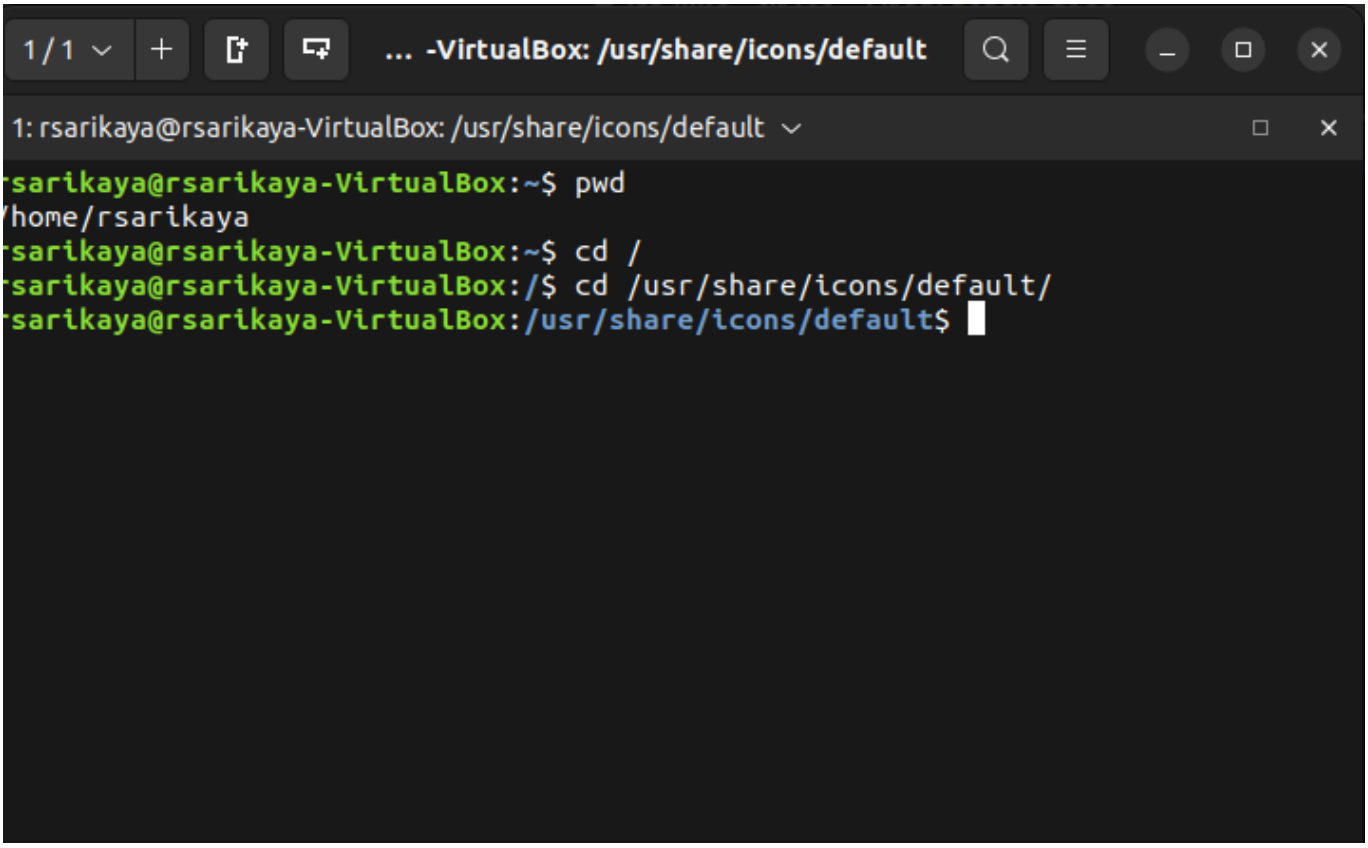
```
rsarikaya@rsarikaya-VirtualBox:~/Downloads$ tree -tug ../lab4files/
[rsarikaya rsarikaya]  ../lab4files/
├── [rsarikaya rsarikaya]  Linux.pdf
├── [rsarikaya rsarikaya]  Tux.png
├── [rsarikaya rsarikaya]  Ubuntu.pdf
├── [rsarikaya rsarikaya]  sample.xls
├── [rsarikaya rsarikaya]  globe.avi
├── [rsarikaya rsarikaya]  document.docx
├── [rsarikaya rsarikaya]  csvfile.csv
├── [rsarikaya rsarikaya]  countries.json
├── [rsarikaya rsarikaya]  website.html
└── [rsarikaya rsarikaya]  helloWorld.c

0 directories, 10 files
rsarikaya@rsarikaya-VirtualBox:~/Downloads$
```

```
1: rsarikaya@rsarikaya-VirtualBox: ~/Downloads
0 directories, 10 files
rsarikaya@rsarikaya-VirtualBox:~/Downloads$ exa ~/lab4files/ -l
.rw-rw-r--  20k rsarikaya 22 Feb 14:16 countries.json
.rw-rw-r-- 284k rsarikaya 22 Feb 14:15 csvfile.csv
.rw-rw-r-- 764k rsarikaya 22 Feb 14:13 document.docx
.rw-rw-r-- 742k rsarikaya 22 Feb 14:12 globe.avi
.rw-rw-r--   76 rsarikaya 22 Feb 14:18 helloWorld.c
.rw-rw-r-- 2.6M rsarikaya 22 Feb 14:10 Linux.pdf
.rw-rw-r-- 8.7k rsarikaya 22 Feb 14:12 sample.xls
.rw-rw-r--  43k rsarikaya 22 Feb 14:10 Tux.png
.rw-rw-r-- 2.1M rsarikaya 22 Feb 14:11 Ubuntu.pdf
.rw-rw-r-- 137k rsarikaya 22 Feb 14:16 website.html
rsarikaya@rsarikaya-VirtualBox:~/Downloads$ exa ~/lab4files/ -lh
Permissions Size User      Date Modified Name
.rw-rw-r--  20k rsarikaya 22 Feb 14:16  countries.json
.rw-rw-r-- 284k rsarikaya 22 Feb 14:15  csvfile.csv
.rw-rw-r-- 764k rsarikaya 22 Feb 14:13  document.docx
.rw-rw-r-- 742k rsarikaya 22 Feb 14:12  globe.avi
.rw-rw-r--  76 rsarikaya 22 Feb 14:18  helloWorld.c
.rw-rw-r-- 2.6M rsarikaya 22 Feb 14:10  Linux.pdf
.rw-rw-r-- 8.7k rsarikaya 22 Feb 14:12  sample.xls
.rw-rw-r--  43k rsarikaya 22 Feb 14:10  Tux.png
.rw-rw-r-- 2.1M rsarikaya 22 Feb 14:11  Ubuntu.pdf
.rw-rw-r-- 137k rsarikaya 22 Feb 14:16  website.html
rsarikaya@rsarikaya-VirtualBox:~/Downloads$
```

```
1: rsarikaya@rsarikaya-VirtualBox: ~/Downloads
rsarikaya@rsarikaya-VirtualBox:~/Downloads$ exa ~/lab4files/ -lh
Permissions Size User      Date Modified Name
.rw-rw-r--  20k rsarikaya 22 Feb 14:16  countries.json
.rw-rw-r-- 284k rsarikaya 22 Feb 14:15  csvfile.csv
.rw-rw-r-- 764k rsarikaya 22 Feb 14:13  document.docx
.rw-rw-r-- 742k rsarikaya 22 Feb 14:12  globe.avi
.rw-rw-r--  76 rsarikaya 22 Feb 14:18  helloWorld.c
.rw-rw-r-- 2.6M rsarikaya 22 Feb 14:10  Linux.pdf
.rw-rw-r-- 8.7k rsarikaya 22 Feb 14:12  sample.xls
.rw-rw-r--  43k rsarikaya 22 Feb 14:10  Tux.png
.rw-rw-r-- 2.1M rsarikaya 22 Feb 14:11  Ubuntu.pdf
.rw-rw-r-- 137k rsarikaya 22 Feb 14:16  website.html
rsarikaya@rsarikaya-VirtualBox:~/Downloads$ exa ~/lab4files/ -lh --no-user -U
Permissions Size Date Created Name
.rw-rw-r--  20k 26 Oct 13:50 countries.json
.rw-rw-r-- 284k 26 Oct 13:50 csvfile.csv
.rw-rw-r-- 764k 26 Oct 13:50 document.docx
.rw-rw-r-- 742k 26 Oct 13:50 globe.avi
.rw-rw-r--  76 26 Oct 13:50 helloWorld.c
.rw-rw-r-- 2.6M 26 Oct 13:50 Linux.pdf
.rw-rw-r-- 8.7k 26 Oct 13:50 sample.xls
.rw-rw-r--  43k 26 Oct 13:50 Tux.png
.rw-rw-r-- 2.1M 26 Oct 13:50 Ubuntu.pdf
.rw-rw-r-- 137k 26 Oct 13:50 website.html
rsarikaya@rsarikaya-VirtualBox:~/Downloads$
```

Question 4 Challenge Question


1 / 1
Tilix: rsarikaya@rsarikaya-VirtualBox: ~
1: rsarikaya@rsarikaya-VirtualBox: ~
rsarikaya@rsarikaya-VirtualBox:~$ pwd
/home/rsarikaya
rsarikaya@rsarikaya-VirtualBox:~$


1 / 1
... -VirtualBox: /usr/share/icons/default
1: rsarikaya@rsarikaya-VirtualBox: /usr/share/icons/default
rsarikaya@rsarikaya-VirtualBox:~$ pwd
/home/rsarikaya
rsarikaya@rsarikaya-VirtualBox:~$ cd /
rsarikaya@rsarikaya-VirtualBox:/$ cd /usr/share/icons/default/
rsarikaya@rsarikaya-VirtualBox:/usr/share/icons/default$

```
1/1 ... rikaya-VirtualBox: /proc/sys/net/ipv4
1: rsarikaya@rsarikaya-VirtualBox: /proc/sys/net/ipv4
rsarikaya@rsarikaya-VirtualBox:~$ pwd
/home/rsarikaya
rsarikaya@rsarikaya-VirtualBox:~$ cd /
rsarikaya@rsarikaya-VirtualBox:/$ cd /usr/share/icons/default/
rsarikaya@rsarikaya-VirtualBox:/usr/share/icons/default$ cd /
rsarikaya@rsarikaya-VirtualBox:/$ cd /proc/sys/net/ipv4
rsarikaya@rsarikaya-VirtualBox:/proc/sys/net/ipv4$
```

```
1/1 Tilix: rsarikaya@rsarikaya-VirtualBox: ~
1: rsarikaya@rsarikaya-VirtualBox: ~
rsarikaya@rsarikaya-VirtualBox:~$ pwd
/home/rsarikaya
rsarikaya@rsarikaya-VirtualBox:~$ cd /
rsarikaya@rsarikaya-VirtualBox:/$ cd /usr/share/icons/default/
rsarikaya@rsarikaya-VirtualBox:/usr/share/icons/default$ cd /
rsarikaya@rsarikaya-VirtualBox:/$ cd /proc/sys/net/ipv4
rsarikaya@rsarikaya-VirtualBox:/proc/sys/net/ipv4$ cd ~/
rsarikaya@rsarikaya-VirtualBox:~$
```

```
1 / 1    ... a-VirtualBox: /usr/share/backgrounds
1: rsarikaya@rsarikaya-VirtualBox: /usr/share/backgrounds
rsarikaya@rsarikaya-VirtualBox:/$ cd /usr/share/icons/default/
rsarikaya@rsarikaya-VirtualBox:/usr/share/icons/default$ cd /
rsarikaya@rsarikaya-VirtualBox:/$ cd /proc/sys/net/ipv4
rsarikaya@rsarikaya-VirtualBox:/proc/sys/net/ipv4$ cd ~/
rsarikaya@rsarikaya-VirtualBox:~$ ls share background
ls: cannot access 'share': No such file or directory
ls: cannot access 'background': No such file or directory
rsarikaya@rsarikaya-VirtualBox:~$ cd /usr/share/backgrounds/
rsarikaya@rsarikaya-VirtualBox:/usr/share/backgrounds$ ls
Blue_flower_by_Elena_Stravoravdi.jpg
canvas_by_roytanck.jpg
Cherry_Tree_in_Lakones_by_elenastravoravdi.jpg
contest
DSC2943_by_kcpru.jpg
Jammy-Jellyfish_WP_4096x2304_Grey.png
jj_dark_by_Hiking93.jpg
jj_light_by_Hiking93.jpg
Mirror_by_Uday_Nakade.jpg
Optical_Fibers_in_Dark_by_Elena_Stravoravdi.jpg
ubuntu2_by_arman1992.jpg
ubuntu_by_arman1992.jpg
ubuntu-default-greyscale-wallpaper.png
warty-final-ubuntu.png
rsarikaya@rsarikaya-VirtualBox:/usr/share/backgrounds$
```

```
rsarikaya@rsarikaya-VirtualBox:/$ cd /home/student
bash: cd: /home/student: No such file or directory
rsarikaya@rsarikaya-VirtualBox:/$ cd /home/rsarikaya/
rsarikaya@rsarikaya-VirtualBox:~$ ls
cis106      Downloads           Music      Pictures         snap
cis106.bk   gitcommands.txt     peru      'pictures peru'   Templates
Desktop     installprograms.sh  picture    practicels       Videos
Documents   lab4files           pictures   Public           website
rsarikaya@rsarikaya-VirtualBox:~$
```

```
rsarikaya@rsarikaya-VirtualBox:/$ cd /home/rsarikaya/
rsarikaya@rsarikaya-VirtualBox:~$ ls
cis106      Downloads           Music      Pictures         snap
cis106.bk   gitcommands.txt     peru      'pictures peru'   Templates
Desktop     installprograms.sh  picture    practicels       Videos
Documents   lab4files           pictures   Public           website
rsarikaya@rsarikaya-VirtualBox:~$ ls Downloads/
lab4files.zip                                Ms_Madi      practicels.tar.xz
Mod3_IntrotoBusiness_GlobalEnvironment.pptx  Ms_Madi.zip
rsarikaya@rsarikaya-VirtualBox:~$
```

```
rsarikaya@rsarikaya-VirtualBox:/dev/disk/by-id$ ls -l
total 0
lrwxrwxrwx 1 root root  9 Nov  3 14:47 ata-VBOX_CD-ROM_VB2-01700376 -> ../../sr0
lrwxrwxrwx 1 root root  9 Nov  3 14:43 ata-VBOX_HARDDISK_VBa6cd9696-8ce11bb0 ->
../../sda
lrwxrwxrwx 1 root root 10 Nov  3 14:43 ata-VBOX_HARDDISK_VBa6cd9696-8ce11bb0-par
t1 -> ../../sda1
lrwxrwxrwx 1 root root 10 Nov  3 14:43 ata-VBOX_HARDDISK_VBa6cd9696-8ce11bb0-par
t2 -> ../../sda2
lrwxrwxrwx 1 root root 10 Nov  3 14:43 ata-VBOX_HARDDISK_VBa6cd9696-8ce11bb0-par
t3 -> ../../sda3
rsarikaya@rsarikaya-VirtualBox:/dev/disk/by-id$
```

```
rsarikaya@rsarikaya-VirtualBox:/dev/disk/by-id$ cd ../../../
rsarikaya@rsarikaya-VirtualBox:/$ cd /usr/share/themes/
rsarikaya@rsarikaya-VirtualBox:/usr/share/themes$ ls -t
Yaru              Yaru-hdpi                 Yaru-purple-dark    Default
Yaru-bark         Yaru-magenta              Yaru-red            Emacs
Yaru-bark-dark    Yaru-magenta-dark         Yaru-red-dark       Raleigh
Yaru-blue         Yaru-olive                Yaru-sage           Adwaita
Yaru-blue-dark    Yaru-olive-dark           Yaru-sage-dark      Adwaita-dark
Yaru-dark         Yaru-prussiangreen        Yaru-viridian       HighContrast
Yaru-dark-hdpi    Yaru-prussiangreen-dark   Yaru-viridian-dark
Yaru-dark-xhdpi   Yaru-purple               Yaru-xhdpi
rsarikaya@rsarikaya-VirtualBox:/usr/share/themes$
```

```
1: rsarikaya@rsarikaya-VirtualBox: /usr/share
rsarikaya@rsarikaya-VirtualBox:/usr/share/themes$ cd ../
rsarikaya@rsarikaya-VirtualBox:/usr/share$ ls -x
accountsservice                 aclocal
acpi-support                    adduser
aisleriot                       alacarte
alsa                            alsa-base
alsa-card-profile               appdata
applications                    apport
apps                            apturl
aspell                          avahi
backgrounds                     base-files
base-passwd                     bash-completion
binfmts                         branding
brltty                          bug
build-essential                 ca-certificates
caffeine-indicator              catdoc
cmake                           cogl
color                           colord
common-licenses                 consolefonts
console-setup                   consoletrans
cowsay                          cups
dbus-1                          dconf
debconf                         debhelper
debianutils                     defaults
desktop-directories             dict
dictionaries-common             distro-info
djvu                            dns
dnsmasq-base                    doc
doc-base                        dpkg
drirc.d                         emacs
emacsen-common                  enchant-2
eog                             evince
evolution-data-server           figlet
file                            file-roller
fish                            flameshot
fontconfig                      fonts
fonts-droid-fallback            fonts-sil-abyssinica
fonts-sil-padauk                foo2qpdl
foo2zjs                         fwupd
gamemode                        games
gcc                             GConf
gdb                             gdm
geany                           gedit
GeoIP                           gettext
ghostscript                     gir-1.0
git-core                        gitweb
glade                           glib-2.0
glvnd                           gnome
gnome-applets                   gnome-background-properties
gnome-bluetooth                 gnome-bluetooth-3.0
gnome-control-center            gnome-initial-setup
gnome-logs                      gnome-mahjongg
gnome-panel                     gnome-remote-desktop
```

```
Applications  Places
1 / 1                                   Tilix: rsarikaya@rsarikaya-VirtualBox: /usr/share
1: rsarikaya@rsarikaya-VirtualBox: /usr/share
netplan                         numpy3
open-vm-tools                   openvpn
orca                            org.gnome.Characters
os-prober                       p11-kit
package-data-downloads          PackageKit
pam                             pam-configs
perl                            perl5
perl-openssl-defaults           pipewire
pixmaps                         pkgconfig
plymouth                        pnm2ppa
pocketsphinx                    polkit-1
poppler                         ppd
ppp                             publicsuffix
pulseaudio                      pyshared
python3                         python-apt
qt5                             readline
remmina                         rhythmbox
rig                             rsync
rsyslog                         rubygems-integration
rygel                           samba
seahorse                        secureboot
sensible-utils                  session-migration
sgml                            sgml-base
sgml-data                       snmp
software-properties             solid
sounds                          source-highlight
speech-dispatcher               ssl-cert
system-config-printer           systemd
systemtap                       tabset
tcltk                           terminfo
tex-common                      texlive
texmf                           themes
thumbnailers                    tilix
totem                           tracker3
tracker3-miners                 transmission
ubuntu                          ubuntu-drivers-common
ubuntu-release-upgrader         ubuntu-wayland
ucf                             ufw
unattended-upgrades             unity
update-manager                  update-notifier
upstart                         usb-creator
usb_modeswitch                  vala
vim                             vlc
vulkan                          w3m
wayland-sessions                X11
xapps-common                    xdg-desktop-portal
xml                             xml-core
xsessions                       yelp
yelp-xsl                        zenity
zoneinfo                        zoneinfo-icu
zsh
rsarikaya@rsarikaya-VirtualBox:/usr/share$
```